

# Participant Handout (Quick Reference Guide)

Cancer Surveillance: Essential Concepts & Practice

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## Part 1: Core Concepts & Reference

### Definition & Purpose

- Surveillance: Systematic, ongoing collection and analysis of cancer data
- Purpose: Measure burden, track trends/equity, guide policy/programs/research
- Global Context: 20M new cases, 9.7M deaths worldwide (2022)

### Surveillance Approaches

| Type     | When to Use                 | Key Consideration           |
|----------|-----------------------------|-----------------------------|
| Passive  | Default population coverage | Variable data quality       |
| Active   | High-quality verification   | Resource-intensive          |
| Sentinel | Rare cancers, rapid insight | Selected sites only         |
| RCA      | Speed-critical applications | Needs e-path infrastructure |

### Case Definition & Coding

- ICD-O-3: Topography (site) + Morphology (histology)
- Staging: AJCC TNM (clinical) + SEER Summary Stage (population)
- Diagnosis Date: Pathology date preferred
- Multiple Primaries: New site/histology rules

### Data Quality Targets

- Completeness: >95% case capture
- Accuracy: ≥98% re-abstraction agreement
- Timeliness: Minimal reporting lag
- Consistency: Pass all NAACCR edits

## **Part 2: Workflow & Standards**

### **Surveillance Workflow**

- I. Case Finding (identify potential cases)
- II. Eligibility Screening (residency, date, reportability)
- III. Abstraction & Coding (ICD-O-3, stage, treatment)
- IV. Quality Checks (edits, re-abstraction audits)
- V. Publication (dashboards, reports)
- VI. Monitor & Improve (feedback loops)

### **Essential Data Sources**

- Clinical: Pathology synoptics, EHR, radiology
- Registry: Hospital-based, population-based (SEER)
- Outcomes: Vital records, patient-reported data
- Ancillary: Claims, pharmacy (validation)

### **Privacy & Security**

- Mandatory Reporting: State public health laws
- Security: Minimum necessary, RBAC, encryption
- Public Outputs: Aggregated data + small-cell suppression

### **Key Standards & Tools**

- NAACCR Volume II: Definitive data dictionary
- ICD-O-3: Global coding standard
- GIS & Analytics: Spatial mapping, predictive modeling

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